Mean-Variance Optimization

Discretionary and Active Portfolio Management

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 8

1 Introduction

Mean-variance optimization remains the cornerstone of modern portfolio theory. The framework provides a systematic approach to balancing expected return against risk, where risk is measured by portfolio variance. In practice, portfolio managers employ mean-variance optimization in two distinct contexts:

1. **Discretionary Portfolio Management**: Maximize absolute return subject to absolute risk constraints
2. **Active or Enhanced Index Management**: Outperform a benchmark while controlling tracking error relative to that benchmark

This document presents the fundamental optimization problems for each context. For discretionary managers, we consider two equivalent formulations: maximizing return subject to a risk constraint, and minimizing risk subject to a return constraint. For active managers, we present analogous formulations in terms of tracking error rather than absolute risk.

2 Notation and Preliminaries

2.1 Basic Definitions

Let:

- N = number of assets in the investment universe
- $\mathbf{w} = (w_1, \ldots, w_N)^T \in \mathbb{R}^N$ = portfolio weights
- $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_N)^T \in \mathbb{R}^N$ = vector of expected returns
- $\boldsymbol{\Sigma} \in \mathbb{R}^{N \times N}$ = covariance matrix of returns (assumed positive semi-definite)
- $\mathbf{1} = (1, 1, \ldots, 1)^T \in \mathbb{R}^N$ = vector of ones

2.2 Portfolio Characteristics

The expected return of a portfolio is:

$$\mu_p = \mathbf{w}^T \boldsymbol{\mu} = \sum_{i=1}^{N} w_i \mu_i \tag{1}$$

The portfolio variance is:

$$\sigma_p^2 = \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i w_j \sigma_{ij} \tag{2}$$

2.3 Portfolio Constraints

Most portfolio optimization problems include the following constraints:

Budget constraint, long-only (fully invested):

$$\mathbf{w}^T \mathbf{1} = \sum_{i=1}^{N} w_i = 1 \tag{3}$$

This constraint is valid only for long-only portfolios.

Budget constraint, long-short (dollar neutral):

$$\mathbf{w}^T \mathbf{1} = \sum_{i=1}^{N} w_i = 0 \tag{4}$$

Long and short positions offset, so the portfolio has zero net dollar exposure.

Long-only constraint (optional):

$$w_i \geq 0 \text{ for } i = 1, \ldots, N \tag{5}$$

Additional constraints may include sector limits, position limits, turnover constraints, etc., but we focus on the fundamental formulations here.

3 Discretionary Portfolio Management

Discretionary portfolio managers seek to construct portfolios that optimize the trade-off between expected return and absolute risk, without reference to a benchmark index. Two equivalent formulations exist: maximizing return for a given level of risk, or minimizing risk for a given level of return.

3.1 Formulation 1: Maximize Return Subject to Risk Constraint

Model

Maximum Return with Bounded Risk

Given a maximum acceptable portfolio variance $\bar{\sigma}^2$, find the portfolio weights that maximize expected return:

$$\begin{aligned} \max_{\mathbf{w}} \quad & \mathbf{w}^T \boldsymbol{\mu} \\ \text{subject to} \quad & \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \leq \bar{\sigma}^2 \\ & \mathbf{w}^T \mathbf{1} = 1 \\ & w_i \geq 0 \text{ for } i = 1, \ldots, N \text{ (optional)} \end{aligned} \tag{6}$$

Interpretation: This formulation is natural when the portfolio manager or investor has a specific risk tolerance $\bar{\sigma}^2$ (e.g., "I can accept at most 15% annual volatility", so $\bar{\sigma}^2 = 0.15^2 = 0.0225$) and wishes to achieve the highest possible return within that risk budget.

This problem has a linear objective and a convex quadratic constraint: a quadratically-constrained quadratic program (QCQP).

3.2 Formulation 2: Minimize Risk Subject to Return Constraint

Model

Minimum Risk with Required Return

Given a minimum acceptable expected return $\underline{\mu}$, find the portfolio weights that minimize portfolio variance:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \\ \text{subject to} \quad & \mathbf{w}^T \boldsymbol{\mu} \geq \underline{\mu} \\ & \mathbf{w}^T \mathbf{1} = 1 \\ & w_i \geq 0 \text{ for } i = 1, \ldots, N \text{ (optional)} \end{aligned} \tag{7}$$

Interpretation: This formulation is natural when the portfolio manager has a return target $\underline{\mu}$ (e.g., "I need to achieve at least 8% expected return") and wishes to minimize the risk taken to achieve that target.

This is a standard quadratic program (QP) with linear constraints, which is efficiently solvable.

3.3 Relationship Between Formulations

The two formulations (6) and (7) are **dual** in the sense that they produce the same optimal portfolios. Specifically:

- For a given variance bound $\bar{\sigma}^2$, solving (6) yields an optimal portfolio with return μ_p^*. Setting $\underline{\mu} = \mu_p^*$ in (7) will yield the same portfolio with variance $(\sigma_p^2)^* \leq \bar{\sigma}^2$.
- Conversely, for a given return target $\underline{\mu}$, solving (7) yields a portfolio with variance $(\sigma_p^2)^*$. Setting $\bar{\sigma}^2 = (\sigma_p^2)^*$ in (6) will yield the same portfolio with return $\mu_p^* \geq \underline{\mu}$.

In practice, (7) is often preferred because it is a convex QP, which is computationally simpler than the QCQP in (6).

3.4 Formulation 3: Risk-Return Trade-off Objective

An alternative approach to mean-variance optimization replaces the explicit risk or return constraints with a single unconstrained objective function that balances risk and return. This formulation directly encodes the investor's risk-return preferences through a risk aversion parameter.

Model

Unconstrained Risk-Return Trade-off

Given a risk aversion parameter $\lambda \geq 0$, find the portfolio weights that maximize the risk-adjusted return:

$$\begin{aligned} \max_{\mathbf{w}} \quad & \mathbf{w}^T \boldsymbol{\mu} - \lambda \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} \\ \text{subject to} \quad & \mathbf{w}^T \mathbf{1} = 1 \\ & w_i \geq 0 \text{ for } i = 1, \ldots, N \text{ (optional)} \end{aligned} \tag{8}$$

Equivalently, this can be written as a minimization problem:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \lambda \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w} - \mathbf{w}^T \boldsymbol{\mu} \\ \text{subject to} \quad & \mathbf{w}^T \mathbf{1} = 1 \\ & w_i \geq 0 \text{ for } i = 1, \ldots, N \text{ (optional)} \end{aligned} \tag{9}$$

Interpretation: The parameter λ represents the investor's risk aversion. A higher λ indicates greater sensitivity to risk, leading to more conservative portfolios. When $\lambda = 0$, the problem reduces to pure return maximization (ignoring risk). As $\lambda \to \infty$, the problem approaches pure risk minimization (the minimum variance portfolio).

3.4.1 Why This Formulation is Often Used in Practice

The risk-return trade-off formulation (8) is widely adopted in practice for several compelling reasons:

1. Computational Simplicity. This is a standard convex quadratic program with only linear equality constraints (and possibly simple bound constraints). Unlike Formulations 1 and 2, there are no inequality constraints on risk or return levels. This makes the problem:

- Faster to solve, especially for large portfolios
- More numerically stable
- Easier to implement with standard QP solvers

2. No Need to Specify Target Levels. Formulations 1 and 2 require the portfolio manager to specify either a variance bound $\bar{\sigma}^2$ or a return target $\underline{\mu}$ a priori. In practice, determining appropriate values for these parameters can be difficult:

- What is the "right" risk target for a given investor?
- What expected return is achievable given current market conditions?

The risk-return trade-off approach sidesteps this issue by asking instead for a relative preference between risk and return (captured by λ), which is often more intuitive.

3. Theoretical Foundation in Expected Utility. The objective function $\mathbf{w}^T \boldsymbol{\mu} - \lambda \mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}$ can be derived from expected utility maximization under the assumption of quadratic utility or normally distributed returns. Specifically, it represents a second-order Taylor approximation to the expected utility $\mathbb{E}[U(R_p)]$ where R_p is the portfolio return. This provides a rigorous economic interpretation: λ corresponds to the investor's coefficient of risk aversion.

4 Active and Enhanced Index Management

Active and enhanced index portfolio managers seek to outperform a benchmark index while controlling the deviation from that benchmark. The relevant risk measure is no longer the portfolio variance σ_p^2, but rather the **tracking error variance**: the variance of the difference between portfolio returns and benchmark returns.

4.1 Benchmark and Tracking Error

Let:

- $\mathbf{w}_B = (w_{B,1}, \ldots, w_{B,N})^T$ = benchmark weights
- $\mu_B = \mathbf{w}_B^T \boldsymbol{\mu}$ = benchmark expected return
- $\mathbf{w}_A = \mathbf{w} - \mathbf{w}_B$ = active weights (deviation from benchmark)

The **expected active return** (or expected excess return over the benchmark) is:

$$\mu_A = \mathbf{w}^T \boldsymbol{\mu} - \mathbf{w}_B^T \boldsymbol{\mu} = (\mathbf{w} - \mathbf{w}_B)^T \boldsymbol{\mu} = \mathbf{w}_A^T \boldsymbol{\mu} \tag{10}$$

The **tracking error variance** is the variance of the active return:

$$\text{TE}^2 = \text{Var}(\mathbf{w}^T\mathbf{r} - \mathbf{w}_B^T\mathbf{r}) = \mathbf{w}_A^T\boldsymbol{\Sigma}\mathbf{w}_A = (\mathbf{w} - \mathbf{w}_B)^T\boldsymbol{\Sigma}(\mathbf{w} - \mathbf{w}_B) \tag{11}$$

4.2 Formulation 4: Maximize Active Return Subject to Tracking Error Variance Constraint

Model

Maximum Active Return with Bounded Tracking Error Variance

Given a maximum acceptable tracking error variance $\overline{\text{TE}}^2$, find the portfolio weights that maximize expected active return:

$$\begin{aligned} \max_{\mathbf{w}} \quad & (\mathbf{w} - \mathbf{w}_B)^T\boldsymbol{\mu} \\ \text{subject to} \quad & (\mathbf{w} - \mathbf{w}_B)^T\boldsymbol{\Sigma}(\mathbf{w} - \mathbf{w}_B) \leq \overline{\text{TE}}^2 \\ & \mathbf{w}^T\mathbf{1} = 1 \\ & w_i \geq 0 \text{ for } i = 1, \ldots, N \text{ (optional)} \end{aligned} \tag{12}$$

Interpretation: This formulation is natural for active managers with a tracking error budget. For example, an enhanced index manager might be constrained to stay within 2% tracking error relative to the S&P 500, and seeks to maximize expected outperformance within that budget.

This is structurally identical to (6), with active return replacing absolute return and tracking error variance replacing portfolio variance.

4.3 Formulation 5: Minimize Tracking Error Variance Subject to Active Return Constraint

Model

Minimum Tracking Error Variance with Required Active Return

Given a minimum acceptable expected active return $\underline{\mu}_A$, find the portfolio weights that minimize tracking error variance:

$$\begin{aligned} \min_{\mathbf{w}} \quad & (\mathbf{w} - \mathbf{w}_B)^T\boldsymbol{\Sigma}(\mathbf{w} - \mathbf{w}_B) \\ \text{subject to} \quad & (\mathbf{w} - \mathbf{w}_B)^T\boldsymbol{\mu} \geq \underline{\mu}_A \\ & \mathbf{w}^T\mathbf{1} = 1 \\ & w_i \geq 0 \text{ for } i = 1, \ldots, N \text{ (optional)} \end{aligned} \tag{13}$$

Interpretation: This formulation is natural when the active manager has a performance target (e.g., "I need to beat the benchmark by at least 50 basis points per year") and wishes to minimize the tracking error variance incurred in achieving that target.

This is a standard QP, analogous to (7).

4.4 Formulation 6: Active Return vs. Tracking Error Trade-off

Just as in the discretionary case, active managers can use an unconstrained formulation that directly balances active return against tracking error variance through a risk aversion parameter.

Model

Unconstrained Active Return vs. Tracking Error Trade-off

Given a risk aversion parameter $\lambda \geq 0$, find the portfolio weights that maximize the risk-adjusted active return:

$$\begin{aligned} \max_{\mathbf{w}} \quad & (\mathbf{w} - \mathbf{w}_B)^T\boldsymbol{\mu} - \lambda(\mathbf{w} - \mathbf{w}_B)^T\boldsymbol{\Sigma}(\mathbf{w} - \mathbf{w}_B) \\ \text{subject to} \quad & \mathbf{w}^T\mathbf{1} = 1 \\ & w_i \geq 0 \text{ for } i = 1, \ldots, N \text{ (optional)} \end{aligned} \tag{14}$$

Equivalently, this can be written as a minimization problem:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \lambda(\mathbf{w}-\mathbf{w}_B)^T\boldsymbol{\Sigma}(\mathbf{w}-\mathbf{w}_B) - (\mathbf{w}-\mathbf{w}_B)^T\boldsymbol{\mu} \\ \text{subject to} \quad & \mathbf{w}^T\mathbf{1} = 1 \\ & w_i \geq 0 \text{ for } i = 1,\ldots,N \text{ (optional)} \end{aligned} \tag{15}$$

Interpretation: The parameter λ represents the active manager's aversion to tracking error. A higher λ indicates greater sensitivity to deviations from the benchmark, leading to portfolios that stay closer to the benchmark. When $\lambda = 0$, the problem reduces to pure active return maximization (ignoring tracking error, which typically results in the benchmark being ignored entirely). As $\lambda \to \infty$, the problem approaches the benchmark portfolio itself (zero tracking error).

5 Summary of Optimization Problems

5.1 Discretionary Management (Portfolio Variance)

Formulation 1	Maximize $\mathbf{w}^T\boldsymbol{\mu}$ subject to $\mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w} \leq \bar{\sigma}^2$
Formulation 2	Minimize $\mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w}$ subject to $\mathbf{w}^T\boldsymbol{\mu} \geq \underline{\mu}$
Formulation 3	Maximize $\mathbf{w}^T\boldsymbol{\mu} - \lambda\mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w}$ (unconstrained)

All formulations include the budget constraint $\mathbf{w}^T\mathbf{1} = 1$ and optional long-only constraints $w_i \geq 0$.

5.2 Active Management (Tracking Error Variance)

Formulation 4	Maximize $(\mathbf{w}-\mathbf{w}_B)^T\boldsymbol{\mu}$ subject to $(\mathbf{w}-\mathbf{w}_B)^T\boldsymbol{\Sigma}(\mathbf{w}-\mathbf{w}_B) \leq \overline{\text{TE}}^2$
Formulation 5	Minimize $(\mathbf{w}-\mathbf{w}_B)^T\boldsymbol{\Sigma}(\mathbf{w}-\mathbf{w}_B)$ subject to $(\mathbf{w}-\mathbf{w}_B)^T\boldsymbol{\mu} \geq \underline{\mu}_A$
Formulation 6	Maximize $(\mathbf{w}-\mathbf{w}_B)^T\boldsymbol{\mu} - \lambda(\mathbf{w}-\mathbf{w}_B)^T\boldsymbol{\Sigma}(\mathbf{w}-\mathbf{w}_B)$ (unconstrained)

All formulations include the budget constraint $\mathbf{w}^T\mathbf{1} = 1$ and optional long-only constraints $w_i \geq 0$.